

6.13. Ethernet Interface

The Ethernet Interface is based on the Intel 82586 chip. Configured in maximum mode, the 82586 can directly address 24-bits of virtual memory. When the 82586 becomes active, it performs data read-write operations in supervisor mode.

The 82586 is connected to the system in a permanent byte-reversed mode, i.e. 82586 bits 0 through 7 are connected to 68000 bits 8 through 15 and vice versa. This causes Ethernet data to be stored in memory in 68010 byte order, whereas 82586 control blocks in memory are byte swapped.

Overall operation of the Ethernet Interface is controlled by the Ethernet control register that can be read or written in the Ethernet page.

Initialization: cleared on all resets			
Interrupts: Level 3			
Reference: Intel 82586 data sheet.			

REGISTER	ADDRESS	DATA	TYPE

CONTROL REG.	0	BYTE	READ/WRITE

The fields of the Ethernet control register are assigned as follows:

ETHERNET CONTROL REGISTER FIELDS		

D0	INT	Interrupt Pending (Read-Only)
D1	ERR	Error Pending (Read-Only)
D2	RES	Reserved
D3	RES	Reserved
D4	INTEN	Interrupt Enable
D5	CA	Channel Attention
D6	LOOPB*	0 => Loopback, 1 => Normal Operation
D7	RESET*	0 => Ethernet Reset, 1 => Normal Operation

INT signals Interrupt from the 82586. *ERR* indicates that a Bus Error occurred during an 82586 channel operation, inhibiting further channel activity. To reset the *ERR* condition, the *RESET* bit in the Ethernet control register must be activated. *INTEN* enables 82586 interrupts to the CPU. *CA* signals channel attention to the 82586. *LOOPB** controls whether the front-end encoder/decoder is configured in loopback mode (*LOOPB** = 0) or connected to the transceiver cable (*LOOPB** = 1). *RESET* initializes the 82586 when active (*RESET** = 0) and allows normal operation when inactive (*RESET** = 1). It also clears the *ERR* condition when active.